

Dual Simplex Method

Steps for dual simplex algorithm:

- **Start with a dual-feasible but primal-infeasible solution:** Start with a basic solution that corresponds to a ready identity, but may violate feasibility.
- **Identify the leaving variable (pivot row):** Choose the basic variable with the **most negative value** (i.e., the most infeasible primal variable) to leave the basis.
- **Find the entering variable (pivot column):** Look at the coefficients in the pivot row. For each non-basic variable with a *negative* coefficient in the pivot row, calculate the ratio: $\theta_{ij} = \left\{ \left| \frac{z_j - c_j}{a_{ij}} \right| : a_{ij} < 0 \right\}$. Choose the variable with the **minimum ratio** to enter the basis.
- **Perform the pivot operation:** Update the tableau by pivoting on the identified element to maintain dual feasibility and improve primal feasibility.
- Check primal feasibility. If all basic variables are non-negative, the solution is primal feasible and optimal. If not, go back to Step 2.

Necessary steps to start a dual simplex table (for maximization type):

- All constraints should be in $\leq$ type.
- The net evaluation $Z_j - C_j \geq 0 \forall j$.
- At least one constraint should contain a negative right hand side.

Ex: Solve the following LPP by dual simplex method:

$$\text{Min } Z = 5x_1 + 6x_2$$

$$\text{Subject to: } x_1 + x_2 \geq 2,$$

$$4x_1 + x_2 \geq 4, \quad x_1, x_2 \geq 0.$$

Solution: To start dual simplex table we rewrite the given problem in the following way:

$$\text{Max } Z' = -5x_1 - 6x_2$$

$$\text{Subject to: } -x_1 - x_2 + s_1 = -2,$$

$$-4x_1 - x_2 + s_2 = -4, \quad x_i, s_i \geq 0.$$

		$C_j \rightarrow$	-5	-6	0	0
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2
0	s_1	-2	-1	-1	1	0
0	s_2	-4 →	-4	-1	0	1
$Z_j - C_j \rightarrow$		$Z = 0$	5 ↑	6	0	0
$\theta_{ij} \rightarrow$			$\left \frac{5}{-4} \right $	$\left \frac{6}{-1} \right $	—	—
0	s_1	-1 →	0	-3/4	1	-1/4
-5	x_1	1	1	1/4	0	-1/4
$Z_j - C_j \rightarrow$			0	19/4	0	5/4 ↑
$\theta_{ij} \rightarrow$			—	$\left \frac{19}{-3} \right $	—	$\left \frac{5}{-1} \right $
0	s_2	4	0	3	-4	1
-5	x_1	2	1	1	-1	0
$Z_j - C_j \rightarrow$			0	0	3	1

Therefore, the optimal solution is:

$$x_1 = 2, \quad x_2 = 0, \quad s_1 = 0,$$

$$s_2 = 4.$$

$$\text{Min } Z = -Z' = -(-10) = 10.$$

Ex: Solve the following LPP by dual simplex method:

$$\begin{aligned} \text{Min } Z &= 2x_1 + 3x_2 \\ \text{Subject to: } 2x_1 + x_2 &\geq 3, \\ x_1 + x_2 &= 2, \quad x_1, x_2 \geq 0. \end{aligned}$$

Solution: To start dual simplex table we rewrite the given problem in the following way:

$$\begin{aligned} \text{Min } Z &= 2x_1 + 3x_2 \\ \text{Subject to: } -2x_1 - x_2 &\leq -3, \\ -x_1 - x_2 &\leq -2, \\ x_1 + x_2 &\leq 2, \quad x_i \geq 0. \end{aligned}$$

The standard form of the above LPP:

$$\begin{aligned} \text{Min } Z &= 2x_1 + 3x_2 \\ \text{Subject to: } -2x_1 - x_2 + s_1 &= -3, \\ -x_1 - x_2 + s_2 &= -2, \\ x_1 + x_2 + s_3 &= 2, \quad x_i, s_i \geq 0. \end{aligned}$$

		$C_j \rightarrow$	2	3	0	0	0
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2	s_3
0	s_1	-3 $\rightarrow$	-2	-1	1	0	0
0	s_2	-2	-1	-1	0	1	0
0	s_3	2	1	1	0	0	1
$Z_j - C_j \rightarrow$		$Z = 0$	-2 $\uparrow$	-3	0	0	0
$\theta_{ij} \rightarrow$			$\left \frac{-2}{-2} \right $	$\left \frac{-3}{-1} \right $	-	-	-
2	x_1	3/2	1	1/2	-1/2	0	0
0	s_2	-1/2 $\rightarrow$	0	-1/2	-1/2	1	0
0	s_3	1/2	0	1/2	1/2	0	1
$Z_j - C_j \rightarrow$			0	-2	-1 $\uparrow$	0	0
$\theta_{ij} \rightarrow$			-	$\left \frac{-2}{-1/2} \right $	$\left \frac{-1}{-1/2} \right $	-	-
2	x_1	2	1	0	1	-1	0
0	s_1	1	0	1	1	-2	0
0	s_3	0	0	0	0	1	1
$Z_j - C_j \rightarrow$			0	-1	0	-2	0

Therefore, the optimal solution is:

$$x_1 = 2, \quad x_2 = 0, \quad s_1 = 1, \\ s_2 = s_3 = 0.$$

$$\text{Min } Z = 4.$$

Special Case:

$$\begin{aligned} \text{Max } Z &= -x_1 \\ \text{Subject to: } x_1 - x_2 &\geq 3, \\ -x_1 + x_2 &\geq 4, \quad x_1, x_2 \geq 0. \end{aligned}$$

		$C_j \rightarrow$	-1	0	0	0
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2
0	s_1	-3	-1	1	1	0
0	s_2	-4 $\rightarrow$	1	-1	0	1
$Z_j - C_j \rightarrow$		$Z = 0$	1	0 $\uparrow$	0	0
$\theta_{ij} \rightarrow$			-	$\left \frac{0}{-1} \right $	-	-
2	s_1	-7 $\rightarrow$	0	0	1	1
0	x_2	4	-1	1	0	-1
$Z_j - C_j \rightarrow$			0	-2	-1 $\uparrow$	0
$\theta_{ij} \rightarrow$			-	-	-	-

We can not calculate the minimum ratio, so none of the basic variables will enter from the last table.

Therefore, the given LPP has infeasible solution.

Exercises

Solve the following LPP by dual simplex method:

1. $\text{Min } Z = 4x_1 + 2x_2$
Sub. To: $x_1 + x_2 = 1,$
 $3x_1 - 2x_2 \geq 2, \quad x_1, x_2 \geq 0.$

2. $\text{Min } Z = 2x_1 + 3x_2$
Sub. To: $x_1 + x_2 \leq 15,$
 $x_1 + 2x_2 \geq 10,$
 $x_1, x_2 \geq 0.$